Aniket Kumar Pradhan

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EDUCATION

Nanyang Technological University (NTU) August 2025 – Expected May 2029
Bachelor of Computing (Honours), Computer Science Singapore

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript
Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face (`huggingface_hub`)
Frameworks & Web: FastAPI, Uvicorn, React, Streamlit, Gradio, Chrome Extensions (Manifest V3)
Databases & Data Systems: MySQL, PyArrow, Parquet, fsspec
Developer Tools: Git, GitHub, pytest, Dev Containers, Google Colab, Jupyter, LaTeX

EXPERIENCE

- **College of Computing & Data Science (CCDS), NTU** October 2026 – Present
  Research Intern | **Advisor:** Prof. Anupam Chattopadhyay
  - Built a Python framework for LLM safety evaluation, implementing residual-harm estimation, bootstrap confidence intervals, and judge-error correction to quantify jailbreak defense effectiveness.
  - Engineered benchmark ingestion across 8 attack families and 13 harm domains, enforcing annotation review and source traceability to prevent unverified data from entering evaluations.
  - Implemented exact statistical estimators to compare 7 filtering budgets from a single 64-response pool, eliminating redundant generation runs for each budget.

- **Undergraduate Research Experience on Campus (URECA), NTU** August 2026 – Present
  Undergraduate Researcher | **Advisor:** Prof. Shuping Dang
  - Developing a deep-learning approach to channel model substitution: a network trained directly on a KL-divergence objective (no labels) that maps intractable lognormal shadowing onto tractable substitute distributions, validated on the lognormal-to-gamma case against numerical baselines.
  - Investigating Tweedie distributions as substitutes: their power parameter nests gamma and inverse-Gaussian, and unlike the lognormal they have a closed-form MGF; preliminary results show 3–9× lower KL divergence than either across 4.3–13 dB shadowing.
  - Derived and proved the exact gamma vs. inverse-Gaussian crossover threshold for KL-optimal substitution, refining a previously approximated value; to be included in a paper in preparation.

PROJECTS

- **Image Signal: Robust AI-Generated Image Detection** | *Python, PyTorch, Computer Vision*
  - Audited standard AI-image benchmarks to uncover container-level data leakage (100% PNG vs. JPEG separability yielding artificial 100% accuracy on zero pixels); built a canonical-decode pipeline closing 12 of 13 metadata leak channels with blocking regression tests.
  - Developed a 19-cell robustness evaluation harness exposing a 0.35 operating-point spread masked by aggregate AUROC (TPR@1%FPR: 0.54 FLUX.1-dev to 0.89 Midjourney across 7 generators); used paired-bootstrap testing to reject training augmentations and CLIP+frequency fusion ablations as statistically insignificant.

- **Sentinel: MARL Perimeter Defense Simulator** | *Python, PyTorch, Reinforcement Learning*
  - Trained a CNN Deep Q-Network (2 Conv2d layers, 3,136 → 256 → 5 head) over 3,000 episodes using smooth-L1 Bellman loss, gradient clipping at norm 10, and a target network synced every 10 episodes.
  - Designed a 5-component reward signal (+10 capture, −50 breach, −0.05 step, etc.) that yielded 4 emergent cooperative behaviors—spatial separation, chokepoint holding, pincer interception, blind spot coverage—with zero hardcoded pathfinding.

- **DICE (Degradation & Image Classification Engine)** | *Python, TensorFlow, OpenCV*
  - Engineered an automated data pipeline using OpenCV to extract, process, and synthetically degrade the VisDrone benchmark dataset with Gaussian blur and sensor noise.
  - Architected and trained a custom 3-block Convolutional Neural Network (CNN) in TensorFlow to perform No-Reference Image Quality Assessment, achieving a 99.77% validation accuracy.

CERTIFICATIONS

- **Deep Learning Specialization** (*DeepLearning.AI*) May 2026
- **Machine Learning Specialization** (*Stanford Online & DeepLearning.AI*) February 2026

CO-CURRICULAR ACTIVITIES

- **Honorary General Secretary** | *NTU UAVionics Student Branch* September 2026 – Present
- **Member, Academics** | *Machine Learning & Data Analytics @ EEE* September 2026 – Present
- **Technology Subcommittee Member** | *IEEE NTU Student Branch* September 2025 – April 2026